

Project ID:

R26-IT-134

1. Topic (12 words max)

IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

2. Research group the project belongs to

AIMS-Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

This IoT-powered smart glasses system tackles a major issue: helping visually impaired people navigate indoors safely and on their own. Without indoor GPS alternatives, they face surprises like obstacles, stairs, ramps, and moving hazards, raising chances of injuries, confusion, and limited movement. Worldwide, over 2.2 billion deal with vision loss, including 43 million blind, making everyday places like homes, offices, and schools tough.[1] Traditional aids like white canes miss high or far-off dangers, and many tech tools need internet, failing in weak-signal spots.

Affected groups include individuals at risk of accidents and reliant on help; families and communities burdened by caregiving and emotional stress; and society facing exclusion, isolation, and high unemployment (up to 70% in some areas). Institutions like schools and workplaces struggle with accessibility, risking inefficiency and legal issues.

User examples: Alex, a 35-year-old blind office worker in urban Sri Lanka, dodges crowds and stairs daily via public transport—the glasses offer offline audio guides for independence. Priya, a 60-year-old caregiver, uses an app for remote alerts, easing worry while juggling work.

The problem's weight: 30-50% higher fall risks hurt quality of life.[2] Impacts span social stigma and depression[3]; technical gaps in tools[7]; economic costs (\$410 billion yearly in lost productivity)[4][5]; ethical inequities; legal non-compliance (e.g., UN disability rights); and environmental strain from extra transport and medical waste. [6][8]

It fits the AIMS research cluster on smart systems. Key industries: healthcare/assistive tech (market growing to billions by 2030), IoT electronics, AI for detection, and education services.[9]

It supports UN SDGs like health (fewer injuries), education access, decent work, reduced inequalities, and sustainable cities through better mobility.[10]

References:

[1] WHO Fact Sheet on Blindness and Visual Impairment:

<https://www.who.int/newsroom/fact-sheets/detail/blindness-and-visual-impairment>

[2] Visual Impairment and Falls Among Older Adults and

Elderly:<https://pmc.ncbi.nlm.nih.gov/articles/PMC9746100/>

[3] Association Between Vision Impairment and Increased Prevalence of Falls in Older

US Adults: <https://agsjournals.onlinelibrary.wiley.com/doi/10.1111/jgs.18879>

[4] Visual Difficulty and Employment Status in the

World:<https://pmc.ncbi.nlm.nih.gov/articles/PMC3917855/>

[5] Global Economic Productivity Losses from Vision Impairment and Blindness:

<https://www.thelancet.com/journals/eclinm/article/PIIS2589-5370%2821%29001322/fulltext>

[6] Eye Health and Economic Development:

<https://www.iapb.org/learn/knowledgehub/elevate/work-and-economic-growth/eye-health-and-economic-development/>

[7] Global Assistive Technologies For Visually Impaired Market:

<https://www.zionmarketresearch.com/report/assistive-technologies-for-visuallyimpaired-market>

[8] The Legal, Ethical, and Psychological Aspects of Self-Determination and Right to Information Access for People with Visual Impairment:

<https://www.nature.com/articles/s41599-024-03798-5>

[9] Assistive Technologies For Visually Impaired Global Market Report 2025:

<https://www.thebusinessresearchcompany.com/report/assistive-technologies-forvisually-impaired-global-market-report>

[10] Going Beyond Health: The Role Eye Care Plays in the SDGs: Discusses eye care's links to SDGs beyond health, including poverty, education, and sustainable cities. :<https://www.sightsavers.org/blogs/2023/12/the-role-eye-care-plays-in-thedgs/>

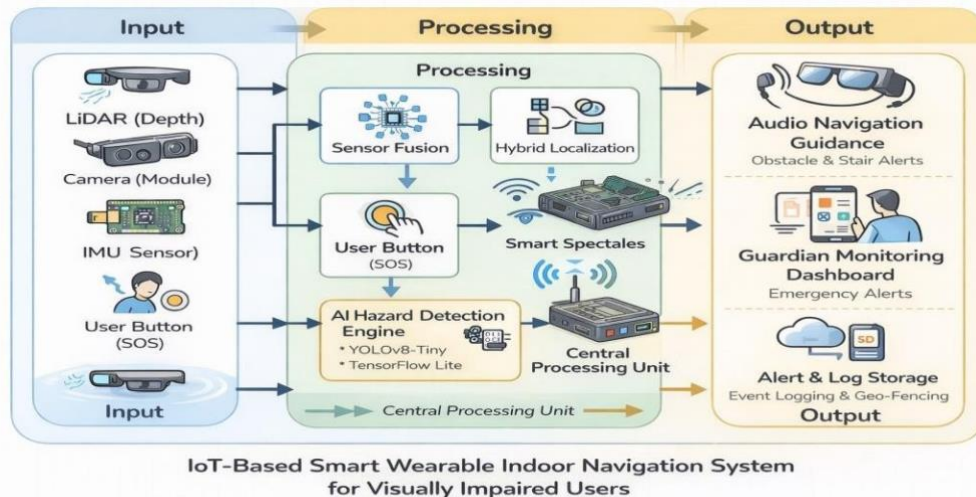
6. Brief description of Existing Research and Systems (300 words max)

Research on tech to help visually impaired folks navigate indoors has come a long way, moving from simple tools like white canes to smarter IoT and AI setups. Since GPS doesn't work well inside buildings, early systems used things like ultrasonic sensors or RFID tags for basic obstacle detection, but they had short range and couldn't predict dangers ahead. In recent years, especially from 2024 to 2026, studies have focused on wearable smart glasses that combine sensors like cameras, LiDAR, and IMUs with on-device AI for real-time hazard spotting—think stairs, furniture, or crowds—without needing constant internet.

For example, one 2025 study developed IoT-enabled glasses with ultrasonic sensors, GPS, and voice alerts to guide users around obstacles and provide navigation cues. Another project created a smart assistive system blending object detection with voice-over tech, aiming to boost independence in daily mobility. Devices like the OrCam MyEye use computer vision for recognizing objects, but they often rely on cloud connections, which can fail offline. Newer prototypes, such as AI-powered smart vision glasses from 2025, incorporate deep learning models like YOLO for quick detection and even vital signs monitoring, cutting fall risks by up to 50%.

However, challenges persist, like making them affordable, private, and reliable in places with poor signals, especially in developing countries like Sri Lanka. Our project builds on these by prioritizing low-power, offline-capable designs for better indoor safety and freedom.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



In response to the problem, the proposed solution is the IoT-based Smart Spectacle System, intended to assist in indoor navigation and safety for blind people. The approach takes a wearable, low-cost, offline-capable solution with real-time sensing, edge computing, and intelligent safety alerts. In contrast to the usual assistive devices in the form of white canes or location-based solutions based on the global positioning system, the proposed solution aims to solve the issues related to indoor navigation.

The proposed system tackles the major shortcomings of the current solutions to the following problems:

- Indoor navigation support availability: The hybrid indoor positioning solution involves dead-reckoning using inertial measurements and vision-based recognition of landmarks for indoor navigation without relying on GPS or the internet.
- Limited hazard detection: Sophisticated sensors like LiDAR, cameras, along with edge AI algorithms (YOLO with TensorFlow Lite), enable early detection of hazards, stairs, and elevation changes.

- Over-reliance on cloud services: Everything necessary is handled on site in edge computing (Raspberry Pi), thus providing reliability even in a cloud setting.
- Lack of adequate safety features: The system includes an integrated safety package with fall detection, SOS distress calls, inactivity alerts, geo-fencing, and a remote guardian feature.
- User discomfort from wearing devices that are too heavy: A split architecture design can lighten the weight of the head-mounted system by moving the heavy computations necessary for the devices into a processing unit located in the pocket of the

The system operates as an integrated pipeline consisting of Input, Process, and Output, ensuring real-time assistance and safety.

Input

- Camera (visual landmarks & obstacle images)
- LiDAR sensor (distance and depth)
- IMU sensor (motion, orientation, fall detection)
- Push button (SOS / mode switching)
- User movement data

Process

- Sensor data collection via smart spectacles
- Wireless transmission (Bluetooth/Wi-Fi) to central processing unit
- Indoor localization using IMU dead reckoning + vision-based landmarks
- AI-based hazard detection using YOLO & TensorFlow Lite
- Safety analysis (fall detection, inactivity monitoring, geo-fencing)
- Decision-making for alerts and navigation guidance

Output

- Real-time audio navigation instructions
- Obstacle and hazard warnings
- Emergency alerts sent to caregivers
- Live status updates on guardian mobile/web platform

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed project focuses on developing an IoT (Internet of Things) based indoor navigation and safety system for people with visual impairments. The project team will acquire the necessary specialized domain expertise through a combination of academic structured self-learning and practical experimentation. Domain knowledge related to assistive technologies for visually impaired users will be gained by reviewing existing research literature, IEEE and ACM publications and case studies on wearable navigation systems, sensor fusion and indoor positioning technologies. In addition, regular consultations with the project supervisor will help validate design decisions and ensure alignment with current research standards.

The skills pertaining to technical know-how that have been identified as required for the project include embedded systems programming, IoT sensor integration, computer vision, edge AI and mobile app development. These skills will be acquired through hands-on experimentation with Raspberry Pi hardware, LiDAR sensors, camera modules and IMU sensors, supported by official documentation and open-source developer communities. The team members will also follow structured online learning resources, including tutorials and courses on OpenCV, TensorFlow Lite and YOLO-based object detection to build the required competency for AI-powered hazard detection. Collaborative development practices within the group will further enable knowledge sharing and skill enhancement across all components of the system.

The data required for the project primarily consists of visual and sensor data collected from indoor environments. Image data and LiDAR distance measurements will be collected using the developed prototype in controlled indoor settings such as corridors, staircases and rooms.

Additional training data for AI models will be sourced from publicly available datasets related to indoor scenes, obstacles and stair detection. All collected data will be anonymized and used solely for research and system evaluation purposes. No personal or sensitive identity data will be stored or shared.

Regarding ethical considerations the project involves assistive technology research for visually impaired users and therefore requires careful ethical awareness. While the system does not collect personally identifiable information or biometric data, ethical clearance may be required if real users are involved in testing or evaluation. In such cases, informed consent will be obtained, and all testing will be conducted in a safe, supervised environment. The project will Page 5 of 13 nest to ensure user safety, updated on adhere to institutional ethical guideli02nd Dec. 2025 privacy and responsible data handling.

9. Objectives and Novelty

Member Name with Registration No	Sub Objective	Task	Novelty
Tharindu R M J - IT22234384	Intelligent Vision Based Danger Detector	• Gather and Prepare Data: Collect a diverse dataset of images showing common indoor hazards (e.g., furniture, stairs, crowds) from public sources or custom captures. Label them accurately with bounding boxes for objects, ensuring variety in lighting, angles, and environments to make the model robust. • Set Up the Model: Choose a lightweight YOLO version (like YOLOv8 nano) suited for edge devices. Configure it with TensorFlow Lite for low-power running on hardware like Raspberry Pi. • Train the Model: Use the labeled images to train the YOLO model on a computer or cloud setup. Finetune hyperparameters like learning rate and epochs to improve detection accuracy, aiming for high precision in spotting and classifying obstacles quickly. tests on new images and real-world videos to check performance metrics (e.g., accuracy, speed, false positives). Simulate indoor scenarios to ensure it predicts hazards early and reliably.	• My key innovation lies in making it truly predictive, not just reactive. While many recent projects use YOLO for basic realtime obstacle spotting in smart glasses, they often stick to simple detection without forecasting risks ahead, like upcoming elevation changes or dynamic crowds in cluttered indoor spaces. Mine goes further by training on a custom dataset tailored for everyday indoor hazards in developing regions like Sri Lanka, where environments can be unpredictable and resources limited. • Another standout is the emphasis on lightweight, offline edge AI. Existing systems might rely on heavier models or partial cloud support, leading to delays or failures in low-connectivity areas. I optimize YOLO (e.g., v8 nano with TensorFlow Lite) for low-power devices like Raspberry

		• Optimize for Deployment: Compress the model to reduce size and power use, making it efficient for wearables. Integrate it with other sensors (LiDAR, IMU) for better context. • Integrate into the System: Link the model to the smart glasses' pipeline— process camera inputs, detect risks, and trigger audio/haptic alerts. Add features like voice notifications for detected obstacles. • Evaluate Overall Impact: Test the full system with users to measure how well it boosts independence and cuts risks, refining based on feedback to meet project goals like SDG alignment and accessibility.	Pi, ensuring quick, private alerts without internet— perfect for real-world use. • Finally, my integration fuses YOLO outputs with other sensors (LiDAR, IMU) for smarter context-aware predictions, reducing false positives and enhancing alerts via audio/haptics. This holistic, user-centered tweak isn't as prominent in other works, which often focus on detection alone. Overall, it boosts independence and safety more effectively for visually impaired folks in underserved spots.
Lakshika V G P - IT22153418	Intelligence Security & Remote Guardian System.	• Mobile App Development: Design and develop a cross-platform mobile application (using Flutter/React Native) for guardians to track the user. • Safety Algorithms: Implement algorithms for Fall Detection and Inactivity Monitoring using accelerometer/gyroscope (IMU) data to detect accidents or medical emergencies. • Geo-fencing Logic: Develop a system to define "Safe Zones" and trigger alerts if the user wanders outside these boundaries.	• Multi-Layered Safety Protocol: Unlike existing systems that only offer basic GPS tracking or simple fall detection, this solution introduces Inactivity Monitoring (detecting unconsciousness where no fall occurs) and Geo-fencing in a single wearable ecosystem. • Voice-Activated Distress Signal: Integration of voice recognition trigger SOS alerts, providing a handsfree safety net for visually impaired users who cannot

		• Cloud Integration: Connect the app with a real-time database (e.g., Firebase) to synchronize location, battery status, and emergency alerts instantly. • SOS Implementation: Integrate the physical Panic Button and Voice-Activated triggers to send immediate SOS notifications.	physically reach the device during an accident. • Context-Aware Monitoring: The dashboard provides guardians with real time environmental context (e.g., "User near stairs") alongside location data, which is rarely found in standard trackers.
Sandaruwan U D H - IT22074690	Smart Spectacle Hardware Module with Multi-Sensory Integration.	• Wearable Hardware Design: Provide a compact and lightweight smart glass structure design for mounting sensors in a way that is comfortable for the wearer to wear while moving indoors. • Sensor Integration: The goal is to integrate the camera, LiDAR, and IMU sensors with the Raspberry Pi so that it can observe its environment in real time • Environmental Data Acquisition: Capture visual data, depth measurements and motion information continuously from the integrated sensors. • Sensor Synchronization: Synchronization of data streams from various	• Spectacle-Based Wearable Design: The system introduces a smart spectacle– based wearable platform that naturally aligns with assistive accessories commonly used by visually impaired individuals, improving usability and acceptance compared to handheld or bulky devices. • Compact Multi-Sensor Integration: The module uniquely integrates a camera, LiDAR sensor and IMU within a single wearable hardware unit, enabling

		sensors in support of real-time, accurate analysis, and navigation decisions. • Edge Processing Support: Prepare and pre-process sensor data for offline, edge based processing, and higher-level AI modules. • User Control Interface: Implement a push-button interface to allow system activation, mode control and emergency interaction. • Audio Output Interface: Provide an audio interface for delivering real-time navigation and hazard alerts to the user.	comprehensive perception of indoor environments including obstacles, staircases and elevation changes. • Depth-Aware Staircase Detection: By incorporating LiDAR-based depth sensing into the spectacle hardware, the system enables early and accurate detection of upward and downward staircases which is not effectively addressed by traditional assistive devices. • Offline Edge Ready Hardware Architecture: The hardware architecture can perform offline edge computing to enable real time processing without cloud connection for reliability in indoor spaces. • Modular and Scalable Design: The proposed hardware architecture is modular allowing future integration of additional sensors/feedback mechanisms without
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			redesigning the entire system.
Aqeel N M – IT22063014	Hybrid Indoor Positioning System with Offline Capability	• This part is the main navigation brain of our smart glasses system. It's designed to figure out exactly where the user is indoors, without needing GPS or any internet connection. • It works by blending two clever approaches: the camera spots everyday landmarks around you—like door frames, walls, or distinctive features—and handles the processing right on the device with simple, efficient vision tools. At the same time, the motion sensors (IMU, using accelerometer and gyroscope) track your steps, turns, and direction changes through a technique called dead reckoning. • By teaming up these two, the system automatically fixes any small drift errors and keeps your position accurate, even in tricky or crowded buildings with no signal. This fully offline setup means the guidance never cuts out, helping visually impaired people move around more freely and confidently on their own. • It also feeds real-time location info straight to the hazard-detection AI and the caregiver monitoring features, so	• Many existing indoor navigation systems for visually impaired people fall short because they depend on pre-installed infrastructure—like BLE beacons, Wi-Fi signals, RFID tags, or UWB setups—which isn't practical in unfamiliar buildings or areas without prior setup, especially in developing regions like Sri Lanka. Pure IMU based approaches suffer from quick positional drift over time, while vision-only systems often need cloud processing or pre mapped landmarks, leading to failures offline or delays in low connectivity spots. Even recent hybrids (from 2024-2026) tend to mix sensors with external aids, raising costs, privacy issues, and reliability problems in real-world varied environments. • What sets this component apart is its truly infrastructure-free, fully offline hybrid design. It fuses lightweight vision-based landmark recognition (spotting everyday features like doors or walls on-the-fly) with IMU dead

		everything in the system connects smoothly.	reckoning, all processed locally on an edge device like Raspberry Pi—no internet, no beacons, no pre-mapping required. This dynamically corrects drift for accurate, uninterrupted tracking in any complex indoor space, making it affordable, private, and portable. It directly feeds real time position data to hazard detection and guardian alerts, empowering greater independence and safety for visually impaired users where other systems can't deliver reliably.
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This part is to be filled by the Supervisor and the Co-supervisor of the Project.

- a. This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b. The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c. The student group can successfully execute the proposed project.

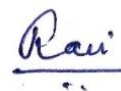
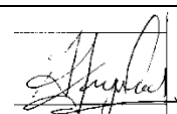
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- d. The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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- e. Any other comments:

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	Title	First Name	Last Name	Signature
Supervisor	Mr.	Ravi	Supunya	
Co-Supervisor	Prof.	Anuradha	Jayakody	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

